DATABASE DESIGN PROJECT

Database Design

From ER Diagram to Normalized Schema

Group 1 - Relational Database Schema (RDBS) & BCNF Analysis

Zhushan He, Selvi Lesmana Putri
Alhagie Kijera, Isaac Ebu-Danso

Methodology

Conversion Rules

From ER Diagram to Relational Schema

Strong Entities

Converted into separate tables with their own primary keys. Each entity becomes a standalone table preserving all attributes.

Examples:

Customer | Product | Order

1:N Relationships

Implemented using Foreign Keys (FK) on the "Many" side of the relationship. The primary key from the "One" side becomes a foreign key.

Example:

Order → customer_id (FK)

M:N Relationships

Converted into Associative Tables (junction tables) with Composite Primary Keys combining PKs from both entities.

Example:

Order ↔ Product

via Order_Items table

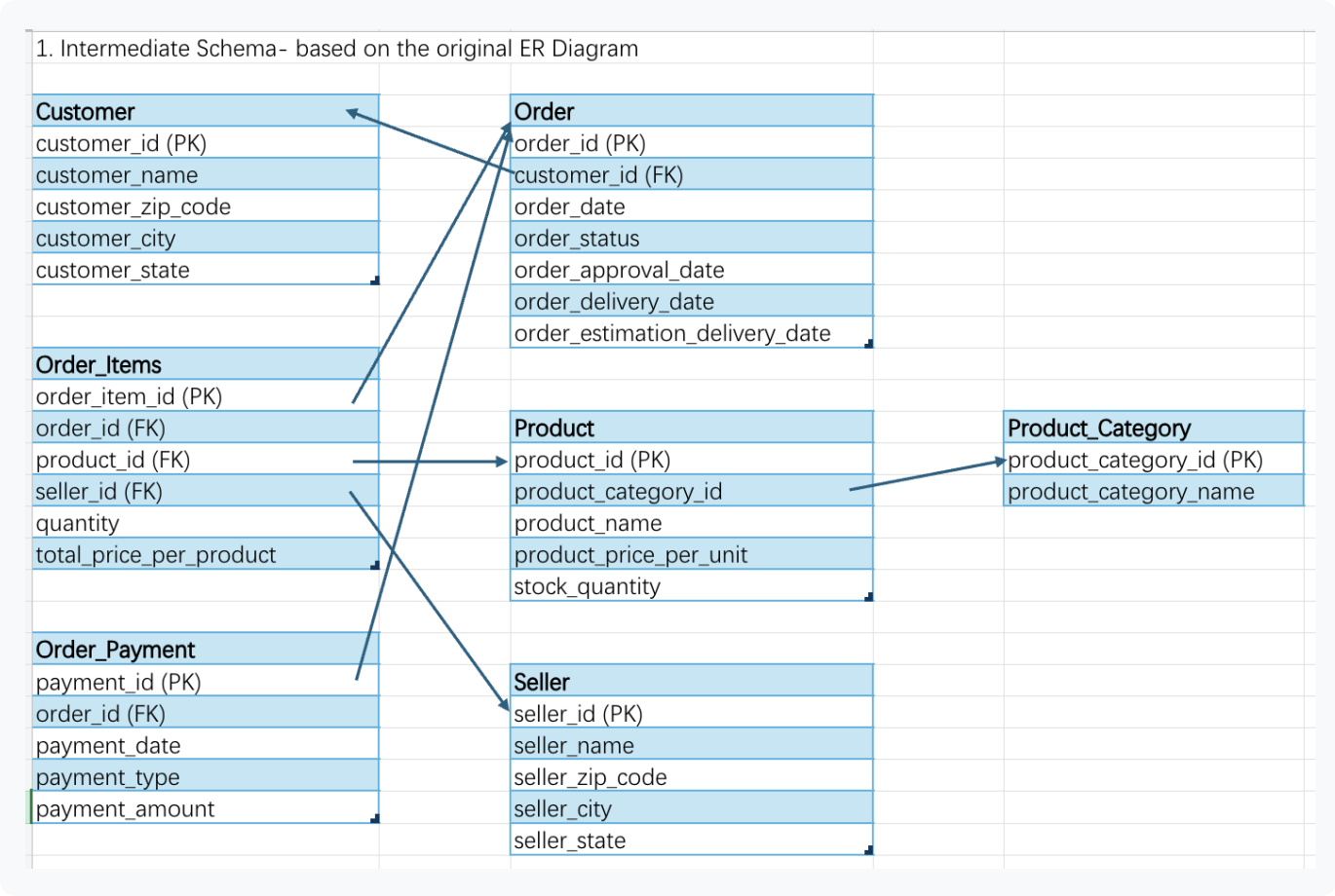
Key Principle

Every relationship in the ER diagram must be translated into a concrete database structure using keys and referential integrity constraints.

Intermediate Relational Schema

Direct Translation from ER Diagram (6 Tables)

Database Schema Visualization



Key Observation

Customer Table: Stores customer information and address including **customer_city** and **customer_state**.

Seller Table: Follows the same structure with location data.

Potential Issue

Data Redundancy: Location information is repeated across multiple tables.

Impact:

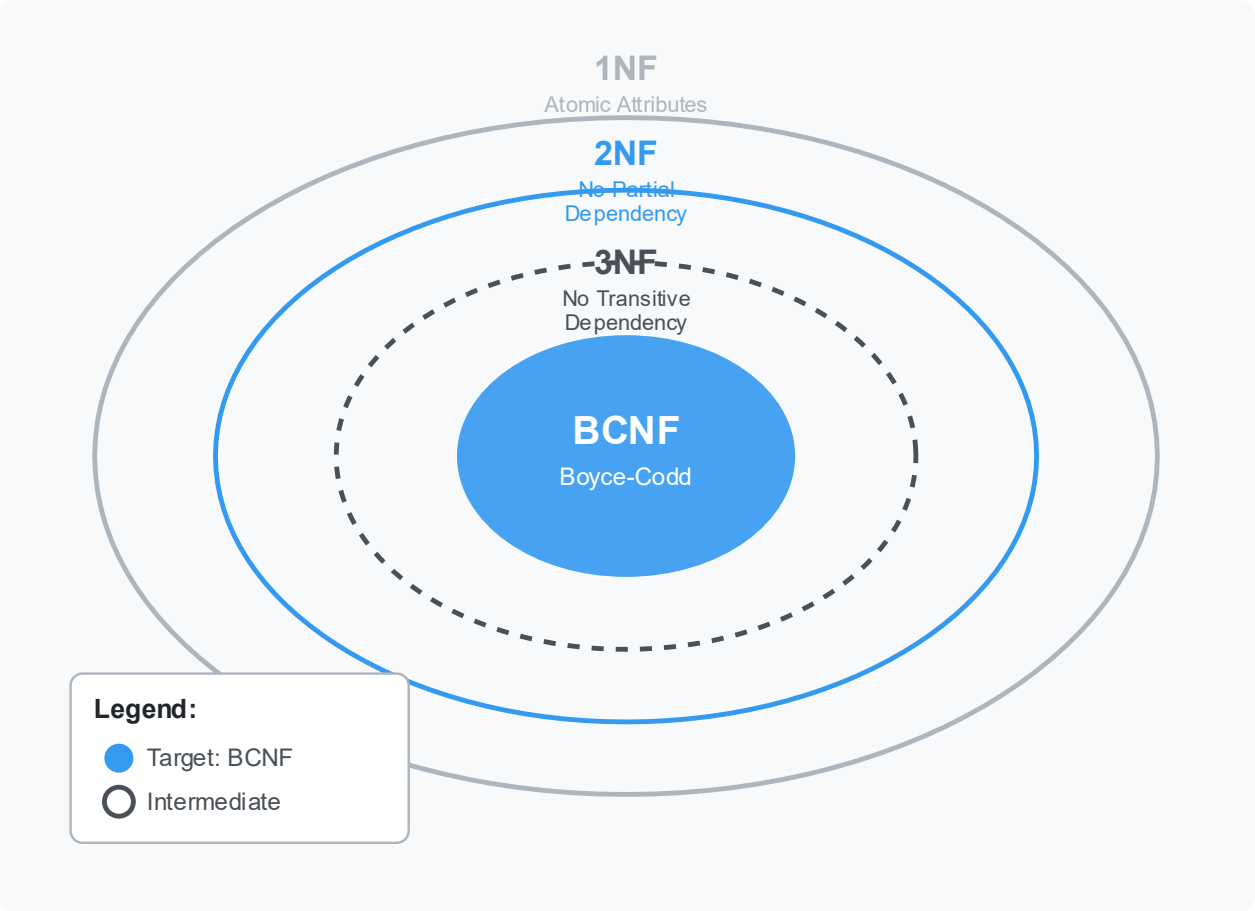
- Update anomalies when city names change
- Inconsistent data across records
- Wasted storage space

Theoretical Framework

Normal Forms

Overview of Normalization Standards

Normal Forms Hierarchy



1 1NF - First Normal Form

Requirement: All attributes must be atomic (indivisible). No repeating groups or arrays allowed.

2 2NF - Second Normal Form

Requirement: Must be in 1NF + No partial dependency. Non-key attributes must depend on the **entire primary key**.

3 3NF - Third Normal Form

Requirement: Must be in 2NF + No transitive dependency. Non-key attributes must depend **only on the key.**

BCNF

Definition: A table is in BCNF if for every functional dependency $X \rightarrow Y$, X is a super key.

Simply put: Every determinant must be a candidate key

Functional Dependency (FD) Analysis

Identifying the BCNF Violation

The Scenario

In the **Customer table**, we have the following structure:

- **PK:** customer_id
- customer_zip_code
- customer_city
- customer_state

The Functional Dependency

customer_zip_code → customer_city, customer_state

Determinant → Dependent

The city and state are **determined by** the zip code, not directly by the customer ID.

The Violation

Issue 1:
City is determined by **Zip_Code** , not by Customer_ID

Issue 2:
Zip_Code is NOT a super key in the Customer table

The Result

Transitive Dependency Detected
Key → Zip → City (City depends on Zip, not directly on Key)

BCNF Violation
The Intermediate Schema **violates BCNF** requirements

Solution Required

Decompose the table to eliminate transitive dependency and achieve BCNF compliance.

Final Relational Schema (Normalized)

Reaching BCNF Compliance (8 Tables)

1 Decomposition

Extracted a new table to isolate location data:

Geolocation

PK: geo_zip_code
Attributes: geo_city, geo_state

2 Cleaning Core Tables

Removed redundant columns and established foreign key relationships:

- → Removed city/state from Customer
- → Removed city/state from Seller
- → Linked via zip_code (FK)

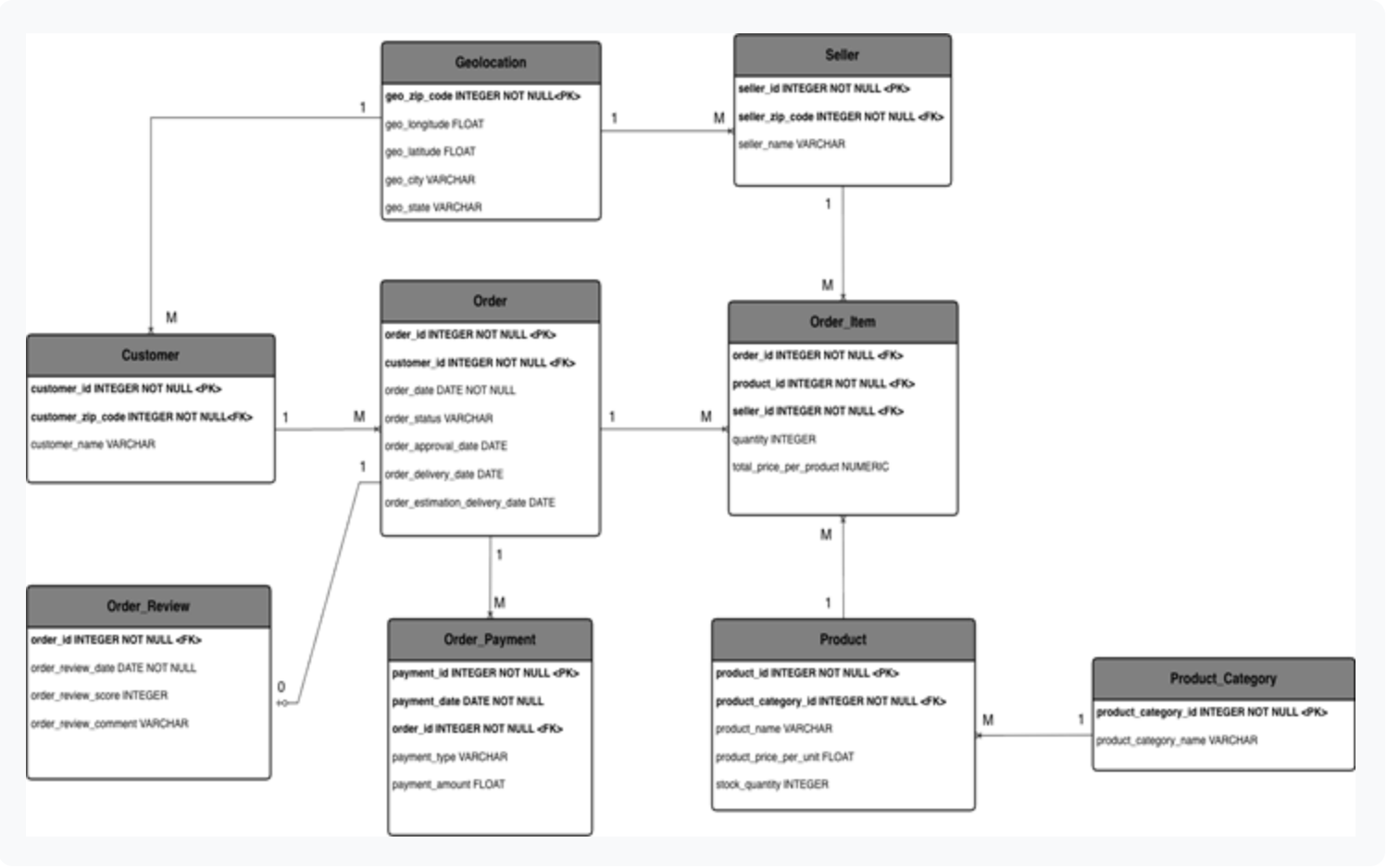
3 Completing the Scope

Added missing entity to complete transaction lifecycle:

Order_Review

Captures customer feedback and ratings

Final BCNF-Compliant Schema (9 Tables)



Critical Thinking & Challenges

Design Decisions & Trade-offs

1 Normalization vs. Performance

Decision

We chose **BCNF** to ensure data integrity and eliminate redundancy. This provides the strongest protection against anomalies.

Trade-off

Our BCNF model assumes a strict **Zip → City** functional dependency for simplicity.

Retrieving a customer's full address now requires a **JOIN operation** between Customer and Geolocation, which is slightly more computationally expensive but safer against anomalies.

2 Entity Classification (Order_Items)

Debate

Is Order_Items a **Weak Entity** or an **Associative Entity**? This classification affects how we model the relationship.

Resolution

Modeled as a **strong associative table** with a Composite PK (order_id, product_id, and seller_id) because it holds critical financial data (price and quantity).

3 Real-World Data Constraints

Issue

In reality, customer does not always provide review on the finished order.

Assumption

Our BCNF model assumes 1:0 cardinality between Order and Order_Review.

CONCLUSION

Project Summary

Transformation

Successfully converted **ER Diagram** to a complete **Relational Schema** following standard conversion rules.

Optimization

Identified and resolved **Transitive Dependencies** in address data through proper normalization to BCNF.

Result

A robust, **BCNF-compliant** database schema with 9 tables ensuring data integrity.

Thank You!